

6 Comparative Analysis of Agentic RAG Frameworks

Table 2 provides a comprehensive comparative analysis of the three architectural frameworks: Traditional RAG, Agentic RAG, and Agentic Document Workflows (ADW). This analysis highlights their respective strengths, weaknesses, and best-fit scenarios, offering valuable insights into their applicability across diverse use cases.

Table 2: Comparative Analysis: Traditional RAG vs Agentic RAG vs Agentic Document Workflows (ADW)

Feature	Traditional RAG	Agentic RAG	Agentic Document Workflows (ADW)
Focus	Isolated retrieval and generation tasks	Multi-agent collaboration and reasoning	Document-centric end-to-end workflows
Context Maintenance	Limited	Enabled through memory modules	Maintains state across multi-step workflows
Dynamic Adaptability	Minimal	High	Tailored to document workflows
Workflow Orchestration	Absent	Orchestrates multi-agent tasks	Integrates multi-step document processing
Use of External Tools/APIs	Basic integration (e.g., retrieval tools)	Extends via tools like APIs and knowledge bases	Deeply integrates business rules and domain-specific tools
Scalability	Limited to small datasets or queries	Scalable for multi-agent systems	Scales for multi-domain enterprise workflows
Complex Reasoning	Basic (e.g., simple Q&A)	Multi-step reasoning with agents	Structured reasoning across documents
Primary Applications	QA systems, knowledge retrieval	Multi-domain knowledge and reasoning	Contract review, invoice processing, claims analysis
Strengths	Simplicity, quick setup	High accuracy, collaborative reasoning	End-to-end automation, domain-specific intelligence
Challenges	Poor contextual understanding	Coordination complexity	Resource overhead, domain standardization

The comparative analysis underscores the evolutionary trajectory from Traditional RAG to Agentic RAG and further to Agentic Document Workflows (ADW). While Traditional RAG offers simplicity and ease of deployment for basic tasks, Agentic RAG introduces enhanced reasoning and scalability through multi-agent collaboration. ADW builds upon these advancements by providing robust, document-centric workflows that facilitate end-to-end automation and integration with domain-specific processes. Understanding the strengths and limitations of each framework is crucial for selecting the most appropriate architecture to meet specific application requirements and operational demands.

7 Applications of Agentic RAG

Agentic Retrieval-Augmented Generation (RAG) systems have demonstrated transformative potential across a variety of domains. By combining real-time data retrieval, generative capabilities, and autonomous decision-making, these systems address complex, dynamic, and multi-modal challenges. This section explores the key applications of Agentic RAG, providing detailed insights into how these systems are shaping industries such as customer support, healthcare, finance, education, legal workflows, and creative industries.

7.1 Customer Support and Virtual Assistants

Agentic RAG systems are revolutionizing customer support by enabling real-time, context-aware query resolution. Traditional chatbots and virtual assistants often rely on static knowledge bases, leading to generic or outdated responses. By contrast, Agentic RAG systems dynamically retrieve the most relevant information, adapt to the user's context, and generate personalized responses.

Use Case: Twitch Ad Sales Enhancement [39]

For instance, Twitch leveraged an agentic workflow with RAG on Amazon Bedrock to streamline ad sales. The system